

Doubly-Fed Induction Generator (DFIG) Power System Dynamic Modeling & Plugin Converter

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Abstract

This document provides a comprehensive overview of the Doubly-Fed Induction Generator (DFIG) wind turbine dynamic model, its implementation using Federico Milano's dq-frame DAE representation, and the architecture of the natID plugin converter developed for the dTwin simulation platform. We outline the physical principles, mathematical formulations, software structure, and step-by-step instructions for testing and running the model on both Linux and Windows environments.

1 Introduction to DFIG Wind Turbines

Doubly-Fed Induction Generators (DFIGs) are widely used in modern wind energy conversion systems due to their ability to operate over a wide range of variable wind speeds while maintaining high efficiency. Unlike traditional synchronous or fixed-speed induction generators, a DFIG wind turbine features a wound-rotor induction machine where:

- The stator windings are directly connected to the three-phase AC power grid.
- The rotor windings are connected to the grid via a back-to-back AC/DC/AC converter (comprising a Rotor-Side Converter, RSC, and a Grid-Side Converter, GSC).

By controlling the rotor currents using vector control techniques, the frequency, magnitude, and phase of the rotor voltage can be adjusted. This allows decoupled control of the active and reactive power injected into the grid. The converter only needs to handle the slip power (typically 25-30% of the total rating), which significantly reduces converter costs compared to full-scale converter wind turbines.

2 Dynamic DAE Model of a DFIG

The dynamic model of the DFIG wind turbine is represented as a system of Differential-Algebraic Equations (DAEs) in the dq-frame, following Federico Milano's standard formulation .

2.1 State Equations (ODEs)

The DFIG has three state variables: the rotor speed ω_m , the q-axis rotor current i_{rq} , and the d-axis rotor current i_{rd} . Their dynamics are governed by:

$$\frac{d\omega_m}{dt} = \frac{\tau_t - \tau_e}{2H_{total}}$$

where H_{total} is the total inertia constant, $\tau_t = P_{opt}/\omega_m$ is the mechanical torque from the turbine using optimal power tracking, and τ_e is the electromagnetic torque.

The rotor current dynamics are modeled using first-order lag equations representing the rotor-side converter controller:

$$\frac{di_{rq}}{dt} = \frac{1}{T_\epsilon} \left(-\frac{x_s + x_\mu}{x_\mu v_h} \left(\frac{P_{opt}}{\omega_m} \right) - i_{rq} \right)$$

where T_ϵ is the controller time constant, x_s is stator reactance, x_μ is mutual reactance, and v_h is the generator bus terminal voltage.

$$\frac{di_{rd}}{dt} = K_V(v_h - v_{ref}) - \frac{v_h}{x_\mu} - i_{rd}$$

where K_V is the voltage regulator gain and v_{ref} is the reference voltage.

2.2 Algebraic Variables & Injections

The electromagnetic torque τ_e is defined algebraically as:

$$\tau_e = -\frac{x_\mu v_h i_{rq}}{\Omega_b(x_s + x_\mu)}$$

where Ω_b is the base frequency. The active and reactive power injected by the DFIG into the grid at the generator bus are:

$$P_h = -\frac{x_\mu v_h i_{rq}}{x_s + x_\mu}$$

$$Q_h = -\frac{x_\mu v_h i_{rd}}{x_s + x_\mu} - \frac{v_h^2}{x_\mu}$$

3 Program Architecture & Conversion Pipeline

The plugin converter parses an IEEE power system XML configuration file (such as 9-bus, 30-bus, 118-bus, or 300-bus systems) and generates a “.dmodl” digital model file and a “.vmodl”.

3.1 ModelConverter Stages

The class `ModelConverter` performs the following pipeline stages:

1. **Stage 1: XML Parsing** – Reads the XML input for buses, branches, generators, DFIG defaults, and standard generator parameters.
2. **Stage 2: DFIG Assignment** – Flags selected generators as DFIG based on the user-configured list (e.g., generator 2) and applies parameter overrides.
3. **Stage 3: Matrix Construction** – Builds `natID` native structures, including the sparse complex admittance matrix (Y-bus) and dense voltage/generator matrices.
4. **Stage 4: Write DMODL Header** – Emits the simulation configurations (startTime, dTime, endTime, maxIter, report).
5. **Stage 5: Variables Section** – Declares state variables along with their initial numeric values.
6. **Stage 6: Parameter Section** – Emits constants, machine parameters, and registers algebraic outputs (`P_e`, `P_h`, `Q_h`) as `[out = true]`.
7. **Stage 7: Equations & Post-Processing** – Emits state equations in `ODEs`: and algebraic output calculations in `PostProc`:

4 Testing & Compilation

4.1 Linux okruženje

To compile the plugin on Linux, navigate to your build folder and run:

```
cmake -build . -config Release
```

To run the automated test suite and convert the IEEE 9, 30, 118, and 300-bus systems:

```
./Release/test_runner /home/abu/SREES/src
```

4.2 Windows Environment

To build on Windows, open Developer Command Prompt for VS:

```
mkdir build && cd build && cmake .. && cmake -build . -config Release
```

Copy the compiled **dfigPlugin.dll** into your dTwin plugins directory. Open dTwin, go to ****Import -> Vjetroelektrana DFIG Converter****, convert your XML file, and solve it!

References

References

- [1] F. Milano, Power System Modelling and Scripting, Springer Science & Business Media, 2010.